# DUNCAN CAMERON-STEINKE

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AI researcher and recent MSc graduate in Mathematics and Statistics, specializing in uncertainty-aware deep learning and volatility forecasting with hidden Markov models.
Three years of experience in data and analytics, most recently as a senior data engineer at ServiceTitan, plus two years as co-founder and CTO of an analytics startup.

## EXPERIENCE

### ASSOCIATE RESEARCHER
Sep 2026 – Present 2 mos

**The University of British Columbia · Kelowna, BC**

- Conducting research in statistics and artificial intelligence, with a focus on uncertainty quantification.
- Developing Belief Distribution Learning (Paper in preparation) with Dr. John Thompson, a novel deep learning framework with learned predictive uncertainty.
- My research poster presented at the Canadian Statistical Society conference is available at https://duncancs.ca/bdl

### SENIOR DATA ENGINEER
Aug 2022 – Apr 2024 1 yr 9 mos

**ServiceTitan**

- Built and owned the data platform on Snowflake and Tableau, managing one engineer.
- Embedded with internal teams to turn open business problems into shipped analytics.
- Architected the data model and owned the pipelines behind a client-facing analytics product (Snowflake, SQL, Python) serving 1,800+ customers.
- Designed an A/B testing dashboard, including sample sizing and significance testing, to measure the impact of new product features.

### ANALYTICS ENGINEER
Jul 2021 – Aug 2022 1 yr 2 mos

**HomeX / Schedule Engine (acquired by ServiceTitan)**

- Established the company's single source of truth for reporting by building the data ingestion, cleansing, and augmentation pipelines for a net new analytics platform.

### CO-FOUNDER & CTO
Jun 2019 – Jul 2021 2 yrs 2 mos

**Backr.ai (YouTube analytics startup)**

- Co-founded a YouTube analytics startup and grew it to 50 customers, including channels with 3M+ subscribers.
- Developed a custom NLP model to classify YouTube comments and analyze sentiment, helping creators grow engagement on their channels.
- Hired and led a team of three full-time developers and a data analyst, owning product roadmap and technical strategy from inception.

Won first prize ($30K) at the DDQIC pitch competition · NextAI Montreal AI accelerator (HEC Montréal, Mila)

### MACHINE LEARNING RESEARCH INTERN
Jan 2019 – Apr 2019 4 mos

**Institute for Infocomm Research · Singapore**

- Authored an NLP research paper (arXiv preprint) on unsupervised text expansion from pre-trained embeddings.
- Improved short-text classification accuracy under limited training data via pseudo-documents.

Research paper (arXiv) (https://arxiv.org/abs/1909.07512) · Singapore International Pre-Graduate Award

## EDUCATION

### THE UNIVERSITY OF BRITISH COLUMBIA
Sep 2024 – Aug 2026

**Master of Science (MSc) · Kelowna, BC**

**Mathematics and Statistics**

GPA: 4.33/4.33

- Coursework in nonparametric models, stochastic processes, differential equations, generalized linear models, and optimization.

### QUEEN'S UNIVERSITY
Sep 2015 – May 2020

**Bachelor of Applied Science (BASc) · Kingston, ON**

**Engineering Physics**

GPA: 3.9/4.0

- Core studies in physics, mathematics, statistics, and computer programming.

Dean's Scholar · H. Jansen Memorial Scholarship (highest standing in Physics I & II)

## CERTIFICATIONS

### MIT SCHWARZMAN COLLEGE OF COMPUTING
2022

**Professional Certificate · Remote**

**Data Science and Machine Learning: Making Data-Driven Decisions**

## RESEARCH PROJECTS

### BELIEF DISTRIBUTION LEARNING
Mar 2026 – Present

**Research paper (in preparation)**

- Created a novel deep learning algorithm grounded in belief theory by combining Subjective Logic with Evidential Deep Learning.
- Introduced per-label uncertainty modeling using Dirichlet distributions.
- Evaluated against published Label Distribution Learning algorithms, demonstrating improved accuracy and better-calibrated uncertainty estimates.

`Python` `TensorFlow` `Keras` `Machine Learning` `Deep Learning` `Uncertainty Quantification`

### VOLATILITY FORECASTING WITH UNCERTAINTY-AWARE COVARIATES
Jul 2025 – Apr 2026

**Research project**

- Added forecasting to the hidden Markov model package hmmTMB.
- Coupled an HMM with an uncertainty-aware deep learning model using a custom loss function.
- Demonstrated improved forecasting of volatility regimes in the TSX composite index compared to HMM or deep learning alone.

`Python` `R` `Hidden Markov Models` `Time Series Forecasting` `Deep Learning`

### FORECASTING RADIO RECOMBINATION LINE SCIENCE
Sep 2025 – Jul 2026

**MSc thesis, UBC Okanagan**

- Scientific research project predicting the performance of a next-generation radio telescope.
- Extrapolated science results given the telescope design and existing surveys as priors.
- Thesis is available at https://duncancs.ca.

`Python` `Statistical Modeling` `Statistics` `Radio Astronomy`

## SKILLS

A/B Testing · AWS · Business Intelligence · Data Engineering · Data Pipelines · Deep Learning · Entrepreneurship · Full Stack · GCP · Hidden Markov Models · Keras · Leadership · Machine Learning · NLP · Product Management · Python · R · Radio Astronomy · Snowflake · SQL · Statistical Modeling · Statistics · Tableau · TensorFlow · Time Series Forecasting · Uncertainty Quantification